

Ideation phase

Define the problem statement

Date	12 July 2026
Team id	
Project name	
Maximum marks	4 marks

Customer problem statement

Farmers often struggle with unpredictable weather, improper irrigation, crop diseases, pest attacks, and declining soil fertility, which reduce crop yield and increase production costs. Many farmers rely on traditional farming methods and lack timely, data-driven guidance for crop selection. They need an affordable, easy-to-use smart agriculture solution that provides real-time recommendations and early alerts to improve productivity, reduce resource wastage, and support sustainable farming.

Section	Smart Agriculture (AI/ML)
I am	A farmer who wants to improve crop yield, reduce farming costs, and use modern technology for better farm management.
I'm trying to	Grow healthy crops, use water and fertilizers efficiently, detect diseases early, and increase productivity and profit.
But	I face unpredictable weather, crop diseases, pest attacks, water shortages, poor soil health, and lack of timely expert guidance.
Because	Traditional farming methods rely on experience rather than real-time data, and I have limited access to advanced agricultural technology and accurate information.
Which makes me feel	Worried about crop failure, financial losses, and uncertain income, while also feeling the need for a simple, reliable, and affordable smart farming solution.

Example



Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Increase crop yield and earn better income	I face unpredictable weather, crop diseases, and inefficient irrigation	I don't have access to real-time data and AI-based recommendations	Worried about crop failure and financial losses
PS-2	A small-scale farmer	Use water, fertilizers, and pesticides efficientl	I waste resources and spend more on farming	Traditional farming methods do not provide accurate monitoring and timely guidance	Stressed about rising costs and low productivity